Round 2 — PathFinder Prototype

Dashboard

Team size 3–5 · teams must be from the same college

Submissions close 31 Aug 2026, 11:59 pm IST · you can edit until then

Sathyabama Institute of Science and Technology, Chennai

Create your team

You'll be the team leader and can invite 2–4 classmates from your college.

Team name

e.g. PathFinders

Create team

AI-Powered Personalized Learning Path Recommender

Design and prototype an AI-powered solution that delivers personalized learning experiences based on an individual's needs, interests, learning patterns and goals.

Background

Online learning platforms offer thousands of courses across diverse domains. While recommendation systems can suggest relevant courses, learners often struggle to identify the right sequence of learning resources needed to achieve a specific goal. Different learners have different skill levels, interests, career aspirations and learning preferences, making a one-size-fits-all approach ineffective. An AI-powered Personalized Learning Path Recommender can bridge this gap by understanding a learner's profile, analyzing learning objectives, identifying skill gaps and generating a structured roadmap of courses, projects and assessments tailored to the individual.

Task

Design and build an intelligent learning assistant that recommends personalized learning paths based on a learner's interests, goals, previous learning history and skill level. The solution should generate a structured learning roadmap, explain recommendations, and adapt suggestions based on user feedback and progress.

What to build

- A conversational interface where learners describe their goals in natural language.
- A learner profiling engine capturing interests, experience level, completed courses and objectives.
- A recommendation engine suggesting relevant courses, projects and learning resources.
- A personalized learning path generator with prerequisites and milestones.
- An AI assistant that explains why each recommendation was made and answers learner queries.
- A dashboard visualizing progress, skill development, milestones and next recommended actions.

Submission guidelines

All five deliverables are required

1 Source Code (ZIP file)

- Submit the complete source code as a ZIP file.
- Include all files required to run the solution.
- Exclude virtual environments, build artifacts and large dependency folders.
- Include a README with setup and execution instructions.

2 Source Code Repository

- Provide a link to the GitHub repo containing the project source code.
- Ensure the repository is accessible to the evaluation team.
- Commit history should reflect the development process.

3 Solution Documentation

- Submit a PDF/PPT covering problem understanding, solution approach, system architecture, AI/ML techniques used, key features and workflows, and challenges faced.

4 Demo Video

- Submit a 3–5 minute demo video.
- Demonstrate the core functionality and key features.
- Explain the overall workflow and user experience.

5 Application Access

- Provide a deployed application URL (if available).
- If not deployed, provide clear instructions for local setup and execution.

Judging criteria

Problem Understanding & Solution Design	20%
Functionality & Feature Completeness	25%
AI/ML Implementation	20%
Innovation & Creativity	15%

User Experience & Interface	10%
Performance & Code Quality	10%